

Cost-Aware Repository Change Localization with Sparse Impact Planning and Bounded Verification

MSc Research Proposal (V1.4, supervisor-ready / print-candidate)

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Abstract

Large language models (LLMs) that edit a software repository must first decide which files a requested change affects. This decision is costly: an explicit file-by-file impact policy over a repository-scale candidate universe spends hundreds of serialized records per call, and a second-stage reasoner that revisits the scope can multiply inference cost by orders of magnitude.

Central identity of the thesis:

This thesis studies whether a cheap sparse file-impact policy can reduce repository-wide localization effort while a bounded second look recovers important files the first pass chose not to change.

We investigate a representation that is cheap by construction — *preserve-by-omission* sparse impact plans — and evaluate it with cheap lexical/static baselines and an explicit *bounded* second-stage omission-recovery mechanism, all under matched inference budgets. Preliminary development evidence shows a large serialization-cost reduction for the sparse representation, a meaningful zero-LLM lexical signal, and — for the first time at candidate level — a robust recovery signal: a frozen BM25+Graph-Neighbor composite ranker (historical label: Classical-CIA) recovers Sparse-omitted missed files above the analytic-Random baseline across the whole budget curve, with task-level bootstrap intervals excluding zero. A clean second-repository DEVELOPMENT run (Saleor, 150 tasks) transfers positively: the frozen composite beats analytic Random on 149 evaluable tasks (B=5 delta +0.231, CI [+0.180,+0.287]), a DEVELOPMENT transfer replication — not a confirmatory result. The confirmatory run on the authorized djangoCMS INTERNAL_TEST (80 tasks, 2026-09-17) ****CONFIRMS**** the fixed-budget recovery signal at every B (B=5 composite 0.165 vs analytic Random 0.028, delta +0.137, CI [+0.075,+0.205]). An identity audit and incremental-evidence ablation show the signal is predominantly lexical (BM25), with the binary graph-neighbor term contributing a small, largely non-significant increment; the thesis therefore does not claim graph novelty. Task-level omission risk is not reliably predictable on current development evidence, so the thesis does not depend on a task-level risk scorer.

1 Background, motivation, and problem

Repository-level coding agents and impact planners must select affected files before editing. Two problems motivate the thesis. First, *impact inference*: did the model identify the right files? Second, *impact-policy representation*: can a complete file-action policy be expressed without spending output budget on hundreds of repeated *preserve* decisions? Our controlled study shows that an explicit full policy at a 16K completion cap serializes on average 144 decision records per candidate universe, while a sparse policy serializes on average 4.9 records (a ~96.6% reduction), with large associated reductions in completion tokens and cost.

Second-stage reasoning is expensive: a graph-guided agent evaluated under a shared protocol consumed on the order of a million tokens per task on the same held-out tasks where our sparse first pass stays below a few thousand tokens. If a cheap first pass already localizes the bulk of affected files, the question is whether a *bounded* second look can recover missed files without paying the full always-on cost.

Given a repository revision P (parent commit), a change request q , and a candidate universe $U(P)$ visible at P , a localization system produces an affected-file set $\hat{S}(P, q)$; the hidden observed change set $S^*(P)$ is used for evaluation only. The observed historical diff is an *observed change-set proxy*, never semantic impact gold (Section 6).

2 Central thesis hypothesis and gap

Hypothesis. A sparse/cheap first-pass localization policy followed by a bounded omission-recovery mechanism can dominate both cheap-localization-only and always-on expensive reasoning on the correctness–cost Pareto frontier, for the repositories and protocols we evaluate.

Everything else supports this story:

- *Preserve-by-Omission* = representation / first-pass contribution;
- *BM25 / classical static change-impact analysis* = cheap baselines;
- *graph-guided agent* = expensive system-level context;
- *task-level omission-risk scorer* = investigated and currently rejected on development evidence;
- *candidate-level bounded verification* = primary next mechanism;
- *second and third repositories* = generalization.

Task-level RiskScorer success is *not* a thesis requirement. Existing work covers several individual components; this thesis evaluates *their combination* as a sparse first pass plus bounded omission recovery under matched budgets (Section 4).

3 Research questions and expected contributions

RQ1. How does sparse explicit impact-policy representation affect inference/output cost and file-selection fidelity?

RQ2. How accurately can cheap lexical/static methods localize impacted files relative to LLM planners under shared protocols?

RQ3. Can bounded second-stage verification recover missed impacted files more efficiently than always-on or random matched-budget verification?

RQ4. How do these correctness–cost trade-offs transfer across repositories and ecosystems?

If task-level risk modeling remains unsupported, RQ3 is answered by the candidate-level mechanism. *Expected contributions:* (1) sparse explicit impact-plan representation with controlled evidence; (2) a reproducible real-commit impact-localization dataset and protocol; (3) fair cheap/classical/LLM/agent comparison under common file-level metrics and explicit budgets; (4) a bounded/selective-compute verification protocol (candidate-level by default); (5) cross-repository correctness–cost evidence; (6) reproducibility artifacts and leakage-control methodology.

No positive algorithmic result is promised.

4 Positioning against related work

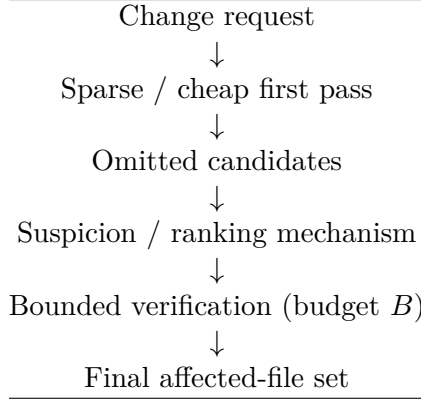
Existing work covers several individual components. This thesis evaluates their combination. Verified related lines (primary sources, 2026-09-16/17):

- *Classical change-impact / ripple effect (RIPPLE line)*. RIPPLE = a known seed $\rightarrow$ recall expansion $\rightarrow$ precision refinement [1]. This thesis differs: an explicit repository-wide sparse policy $\rightarrow$ a rejected residual set $\rightarrow$ bounded reconsideration of rejected candidates.
- *Change-Patterns Mapping* (TSE 2022) [2]: boosts/reranks the initial impact set from historical patterns. This thesis allocates *new inspection to the excluded residual set* rather than reranking the selected set.
- *ArtifactSync*: a demo-level mechanistic analogue (uncertainty- driven deeper context) but a different post-commit synchronization task, not file-impact localization.
- *Modern transform + program-dependence-graph impact analysis* [3]: graph+LLM impact analysis is active; our graph is one optional verifier.
- *Selective-compute / efficiency+verification priors* [4, 5]: selective retrieval / efficient verification at repository level for code-completion/generation tasks, guarding generic “selective escalation is novel” claims.
- *Repository-level coding and retrieval agents* [6, 7, 8, 9, 10]: LLM agents and retrieval-conditioned generation over whole repositories; this thesis evaluates a cheap sparse first pass against shared-cost context.
- *Graph-guided issue localization* [11, 12, 13]: expensive graph-guided competitors (LocAgent, GraphLocator, LLM-driven iterative code-graph search).
- *Classical / history-learned change-impact predictors* [14, 15, 16, 17]: dependency- and history-based impact prediction and industrial studies.

Graph, history, and selective computation are *not* novelty claims by themselves. The comparison matrix (Section 7) records whether each line has an explicit file policy, sparse omission representation, cheap first pass, omission recovery, hard/matched budget, real-commit evaluation, and cross-repository evidence. Where novelty is only defensible once experiments support it, it is marked *candidate novelty — not yet claimed*.

5 Proposed two-stage architecture

1. **Sparse/cheap first-pass impact policy**: explicit file-level policy with preserve-by-omission serialization.
2. **Candidate-level bounded verification** (primary): rank the first-pass omitted candidates with independent structural / retrieval / history evidence, and verify only those under a hard per-task budget B . Compare with always-verify and random matched-budget verification.



6 Dataset construction, labels, and leakage controls

- **Real history, parent-only inputs.** Inference uses the parent commit P ; the target T is hidden. Live HEAD is never used.
- **Eligibility and exclusions.** Single-parent commits with a meaningful intent, touching only production source files; tests, configurations, docs, migrations, and generated files are excluded; intent-path leakage is a hard exclusion.
- **Deduplication.** Exact-proxy-set / shared-reference / suspected- related changes are removed with frozen deterministic rules.
- **Observed change-set proxy.** $P \rightarrow T$ changed production files are an *observed* proxy, never semantic impact gold. Localization claims are scoped to historical-file recovery unless semantic adjudication supports more.
- **Splits.** Metadata-only, deterministic split frozen before any model result. For djangoCMS V2-LARGE: development, an untouched internal test, and a reserve. The v1 40 are LEGACY-EXPOSED and never reused as untouched confirmation.

7 Experimental design, baselines, budgets, metrics, statistics

7.1 Budget curve, not a single point

The primary scientific object is the *budget curve* $B \in \{0, 1, 3, 5, 10\}$; $B=0$ is Sparse alone. $B=5$ is a convenient reference operating point, not a scientifically privileged single claim. We also report the normalized sensitivity $B/|\text{omitted set}|$ as a secondary view.

7.2 Analytic Random control

For ranking-only experiments the Random control is the analytic hypergeometric expectation: for a task with N_t omitted candidates, M_t Sparse-observed FNs, and budget B , $\mathbb{E}[X_t] = B \cdot M_t / N_t$ (with B clipped to N_t). The comparison is per-task observed recovery vs this analytic expectation, aggregated with task-level bootstrap intervals. Random repetitions are never independent tasks.

7.3 Primary endpoint

False-Negative Recovery Rate / Omission Recovery Rate @ B : per task, recovered Sparse-observed FNs within B / all Sparse-observed FNs. Secondary: absolute recall gain, FNR reduction, fraction of Oracle@ B gap closed, candidates inspected per recovered FN, precision/F1 (safety metrics), runtime, index/build cost.

7.4 Anchors (per B)

Sparse/ $B=0$; Analytic Random; BM25; Path-token; Graph/static; BM25+Graph-Neighbor Composite (historical label: Classical-CIA; the frozen primary); history/co-change (where parent-visible history is available); one fixed simple hybrid (0.5 BM25 + 0.5 graph neighbor, a rank-equivalent redundant control — confirmed by the ranker-identity audit, not an independent baseline); Oracle (ranking headroom); InspectAll (exhaustive). $Oracle@B \neq InspectAll$: Oracle is ranking headroom; InspectAll is exhaustive reconsideration.

7.5 Experimental Design at a Glance

Study	Purpose	Repo / Split	Method	Unit	Budget / Reps	Primary metrics	Status
Controlled	sparse-vs-full cost	djangoCMS / controlled	Full vs Sparse	task	cap 16K / 3 reps	tokens, records, cost, P/R/F1	Done
Historical	selection fidelity	djangoCMS / held-out	Full vs Sparse	task	cap 16K / 3 reps	P/R/F1/FNR	Done
Agent	expensive context	djangoCMS / held-out	graph-guided agent	task	per-call	P/R/F1, cost	Done
Cheap baselines	zero-LLM signal	djangoCMS / dev	BM25 / graph	task	K curve	P/R/F1	Done
Task risk	risk predictability	djangoCMS / dev	features	task	3 reps	AUROC/AUPRC	Negative / closed
Candidate recovery	omission recovery	djangoCMS / dev	composite ranker + verifier	cand.	$B \in \{0..10\}$	FN-recovery	DEV replicated; confirmatory pending
Cross-repo transfer	transfer	Saleor / dev	frozen method	task	B curve	per-repo	Saleor DEV replication complete

Table 1: Experimental Design at a Glance (internal IDs appear in traceability notes only). Status as of 2026-09-17.

7.6 Statistics and multiple-comparison discipline

Task is the independent unit; candidate rows are nested. One primary endpoint (recovery@ B); one primary arm-vs-Random comparison; secondary endpoints labeled; task-level paired bootstrap; no post-hoc feature-family winner; exploratory analyses reported as exploratory.

8 Preliminary evidence

All numbers are from audited studies in this repository; they are development or controlled evidence with their bounds.

- **Representation effect (controlled).** At a 16K completion cap, Sparse serialized on average **4.9** decision records vs **144.0** for Full (**~96.6%** reduction), with large completion-token and cost reductions. No universal semantic-superiority claim.

- **Historical real-commit study.** 60 cells over 10 held-out real djangoCMS commits: selection is difficult, FNR is high, and there is no Full-vs-Sparse superiority claim (paired deltas' intervals cross zero). Sparse is substantially cheaper.
- **Graph-guided agent comparison.** System-level shared-protocol context: 402 calls / 32.8M tokens / $\sim \$9.93$ estimated on 10 tasks, with 5/10 non-usable outcomes (2 timeout + 1 context-length + 2 completed-but-empty). Not a reproduction of the published fine-tuned result.
- **Cheap/static development baselines.** BM25 provides a meaningful zero-LLM signal; graph-ranked seeds are comparable; a classical CIA (dependency propagation) baseline is implemented on 150 development cases.
- **Task-level omission-risk investigation.** Live Sparse development inference (90 cells; then 431 calls with a fail-closed stop at the 2.5M-token ceiling). A task-level RiskScorer is not justified: the v1 signal does not replicate within the new development cohort and the pooled signal is a candidate-universe-size artifact.
- **Candidate-level bounded verification (Route B, development).** A frozen BM25+Graph-Neighbor composite ranker (historical label: Classical-CIA; exact formula = normalized BM25 + binary graph-neighbor, per the ranker-identity audit) recovers Sparse-omitted missed files above the analytic-Random baseline across the whole budget curve $B \in \{1, 3, 5, 10\}$, with task-level bootstrap intervals excluding zero at every point (e.g. at $B=5$: pooled macro recovery 0.163 vs random 0.029, interval $[+0.09, +0.18]$), 5/5 development folds positive, and no omitted/universe-size artifact. A small verifier pilot (30 calls, $\sim \$0.002$) found the frozen ranker places the recoverable missed files within the top- B (Oracle-in-top- $B \approx 1.0$) and the verifier recovers ~ 0.86 – 1.0 of them, so the dominant loss is first-pass omission.
- **Saleor DEVELOPMENT transfer replication (2026-09-17).** A clean Saleor DEVELOPMENT Sparse run (150 tasks $\times$ 3 reps = 450 cells; 446 valid / 4 failed; 7,316,986 tokens / $\sim \$2.31$; 0 truncations; no Saleor-specific tuning) enabled a frozen Route-B transfer test on 149 evaluable tasks: the composite beats analytic Random at every B ($B=5$: 0.237 vs 0.006, delta +0.231, CI $[+0.180, +0.287]$; 5/5 folds positive; no size artifact). This is a **DEVELOPMENT transfer replication, NOT a confirmatory result**; Saleor INTERNAL_TEST/RESERVE remain sealed.
- **djangoCMS confirmatory result (2026-09-17, CONFIRMS).** The frozen confirmatory protocol ran on the authorized djangoCMS V2 INTERNAL_TEST (80 tasks) under the frozen V2 packet and budget (560 calls / 1,470,174 tokens / $\sim \$0.51$; 0 failures; 0 excluded): the composite beats analytic Random at every B ($B=1$ 0.059 vs 0.006; $B=3$ 0.110 vs 0.017; $B=5$ 0.165 vs 0.028, delta +0.137, CI $[+0.075, +0.205]$; $B=10$ 0.267 vs 0.055), with task-level bootstrap intervals excluding zero at every B and no size artifact. ****Fixed-budget Route B is scientifically CONFIRMED on djangoCMS INTERNAL_TEST.**** The confirmatory test is permanently used and will not be reused for adaptive-budget (P2) selection.
- **Incremental-evidence ablation (zero API, 2026-09-17).** Composite-minus-BM25 paired task-level deltas are small with bootstrap CIs including zero at most B on both repositories (djangoCMS $B=5$ +0.003 $[-0.015, +0.019]$; Saleor $B=5$ +0.003 $[-0.017, +0.024]$); the replicated cross-repo signal is **predominantly lexical (BM25)**. The thesis does not claim graph novelty.

- **Classical/history evidence.** A parent-visible history/co-change arm also beats analytic Random where history is available; the BM25+Graph-Neighbor composite remains the frozen primary.

These results motivate a larger development dataset, protection of an untouched internal test, and the bounded-verification design; they are not rewritten as a success.

9 Cross-repository generalization

Stage 1 djangoCMS V2-LARGE (within-repo large evidence + untouched internal test) → Stage 2 Saleor (different real system, same Python/Django stack) → Stage 3 NestJS (cross-language / cross-framework evidence). Each stage states its claim boundary. NestJS is preferred but not forced; a predeclared eligible-pool yield criterion and a backup protocol are frozen.

9.1 Saleor DEVELOPMENT transfer replication (completed, 2026-09-17)

The Saleor DEVELOPMENT sparse run ($150 \times 3 = 450$ cells; 446 valid / 4 failed; 7,316,986 tokens / $\sim \$2.31$; 0 truncations; no Saleor-specific tuning) and the frozen Route-B transfer test (149 evaluable tasks) replicate the bounded omission-recovery signal on a second, much larger repository ($B=5$ composite 0.237 vs analytic Random 0.006; delta +0.231, CI [+0.180, +0.287]; 5/5 folds positive; no size artifact). This is **DEVELOPMENT transfer replication only** — **NOT confirmatory**. Saleor INTERNAL_TEST/RESERVE are sealed and were never opened. Per-repository primary; no djangoCMS+Saleor pooling as the headline.

9.2 djangoCMS confirmatory result (2026-09-17, CONFIRMS)

The frozen confirmatory protocol ran on the authorized djangoCMS V2 INTERNAL_TEST (80 tasks) under the frozen V2 packet and budget (560 calls / 1,470,174 tokens / $\sim \$0.51$; 0 failures; 0 excluded): the frozen BM25+Graph-Neighbor Composite recovers Sparse-omitted historically-changed files above the analytic Random control across the whole budget curve ($B=1$ 0.059 vs 0.006; $B=3$ 0.110 vs 0.017; $B=5$ 0.165 vs 0.028; $B=10$ 0.267 vs 0.055), with task-level bootstrap intervals excluding zero at every B and no size artifact. ****The fixed-budget Route-B result is scientifically CONFIRMED on djangoCMS INTERNAL_TEST.**** The confirmatory test is permanently used and will not be reused for adaptive-budget (P2) selection; P2 is evaluated on development data only, with Saleor INTERNAL_TEST sealed as a possible future P2 confirmation set.

9.3 Scope contract

CORE MSc: Preserve-by-Omission representation evidence; fixed-budget Route-B development + the confirmed djangoCMS INTERNAL_TEST result; semantic-proxy audit; djangoCMS; Saleor as secondary replication if completed. *Conditional / Future:* adaptive B_t (P2, development-data exploration, Nov 2026 – Mar 2027); learned budget allocation; NestJS cross-language; end-to-end regeneration correctness; persistent memory; richer causal-graph verifier.

10 Threats, contingency, and timeline

10.1 Threats to validity

Internal: the observed proxy is not semantic gold; leakage control is mandatory; no hidden-test peeking. External: n and single-repository generalization; V2 and cross-repository evidence are the mitigation, with per-repository primary metrics. Statistical: low negatives, multiple

comparisons, wide random bands — addressed by pre-registration, class-balance gates, analytic-Random controls, and bootstrap uncertainty. Representation: LLM serialization cost spans providers; both recorded and frozen-pricing cost are reported.

10.2 Semantic-proxy validity

Because the historical diff is only a proxy, we prepare a stratified development-only sample with evidence packets for later human adjudication.

The protocol is `docs/SEMANTIC_PROXY_AUDIT_PROTOCOL.md`.

Machine-assisted notes are never semantic gold.

10.3 Contingency

If candidate-level bounded verification does not beat analytic Random at the frozen budget, the negative result is frozen and reported; no complex verifier is invented.

10.4 Future work: adaptive / cost-sensitive verification allocation

Inspired by the Shichao Zhang adaptive-resource line (Learning- k , one-step KNN, cost-sensitive KNN, adaptive-neighborhood) [18, 19, 20, 21] and selective-prediction / learning-to-defer literature, we sketch a future adaptive budget: $B_t^* = \arg \min_B [C_{\text{verify}}(B) + \rho \mathbb{E}[\text{remaining omission loss} \mid \text{evidence}, B]]$, with ρ a sensitivity, not a claimed monetary truth. This is NOT yet a contribution and is NOT implemented before fixed-budget Route B is established (status: CONDITIONAL; see the P2 pre-registration note).

10.5 Timeline (2026-11 to 2027-10)

Intended substantive research/thesis completion: **2027-07/08**. The 2027-09/10 window is a publication, revision, and administrative buffer. No experiments are fabricated merely to fill months. P2 (adaptive budget) is a conditional exploration: if it does not justify a candidate on development data, it is closed as negative and the fixed Route-B result (already confirmed on `djangoCMS INTERNAL_TEST`) remains the thesis basis. NestJS is conditional on its frozen gates and evidence.

11 References

All entries below were verified against primary sources (arXiv API / Crossref) on 2026-09-16/17. Preprints are explicitly marked. No fabricated or placeholder citations remain; the source bibliography is `references.bib` (V1.4).

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Window	Planned work
2026-11	Fixed Route-B confirmation (djangoCMS INTERNAL_TEST; CONFIRMED 2026-09-17) + P2 literature/formulation freeze
2026-12	Simple Shichao-inspired adaptive-budget policies (score-gap, marginal-score, cost-ratio stopping) on development data
2027-01	Cost-sensitive / one-step / demand-driven adaptive-budget analogues (development data only)
2027-02	Newly discovered competitor/alternative algorithm reproduction (selective/cascade/optimal-stopping families)
2027-03	P2 comparison/freeze/pre-registration on development; negative closure if not justified; Saleor INTERNAL_TEST only if pre-registered
2027-04	Optional external-validity extension (NestJS or predeclared backup if suitability gate passes); no forced cross-language claim
2027-05	Consolidated statistical analysis; Pareto/frontier analysis; semantic-audit integration; final threat-to-validity revision
2027-06	Thesis chapters/results writing; artifact cleanup; reproducibility package; supervisor review round 1
2027-07	Thesis full draft; corrections; final experimental closure; submission-ready research manuscript draft
2027-08	Target research/thesis completion; supervisor review round 2; final figures/tables/appendices; defense/seminar preparation
2027-09	Publication revision/submission window; reviewer-style internal audit; contingency for delayed replication or manuscript resubmission
2027-10	Final buffer: journal/conference resubmission, thesis formatting/administrative closure, archival artifact release

Table 2: Work schedule, November 2026 – October 2027. July–August 2027 is the intended substantive research/thesis completion; September–October 2027 is an honest publication/revision/administrative buffer, not fabricated work. November 2026 – March 2027 is the P2 adaptive-budget exploration program (development data only).

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